



Locating Places on Earth



Learning Outcomes

The learner:

- explains what latitudes and longitudes are.
- distinguishes between latitudes and longitudes and elaborates on their features.
- uses latitudes and longitudes to locate places on a map.

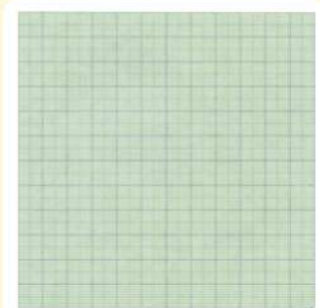


Let's Take-off

These lines are called ad.
I have seen lines like these on g..... and
m.....
These lines do not actually exist on the

Communication

21st
CENTURY
SKILLS



Look at the picture of this globe. It has a similar criss-cross of lines. The lines drawn on maps and globes are real lines, but the surface of the earth does not have them. We imagine they are there and therefore these lines are imaginary lines. They have special names in geography. These lines are called **latitudes** and **longitudes**.

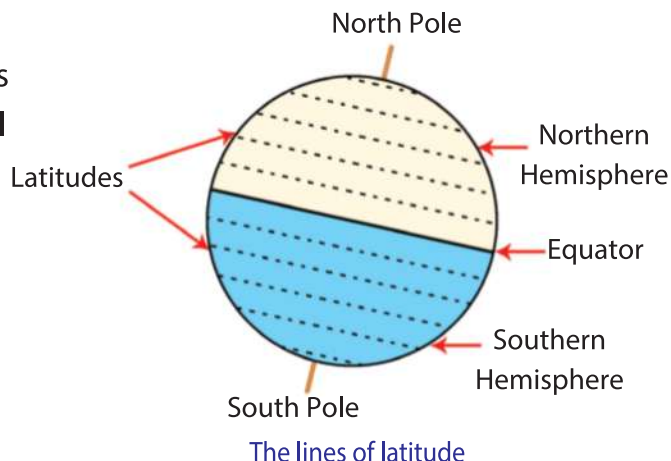
LATITUDES

We know that the equator is drawn on maps and globes around the centre of the earth. It is a circle that divides the earth into two equal halves called **hemispheres** (hemi= half) which are north and south of the equator.



Lines are drawn parallel to the equator in both these hemispheres. The lines are circles from the equator to the poles. These parallel lines, including the equator are called latitudes.

As the latitudes move closer to the poles, the circles get smaller. Eventually, at each pole, the latitude is just a dot! The diagram shows you all these latitudes.



Each latitude is given a number. The equator is numbered as 0 degree, which is written as 0° . The numbers of the other latitudes go from 1° to 90° in each hemisphere—north and south. They have a number with the letters 'N' or 'S' written after it. This tells us whether the latitude is in the northern hemisphere (N), or in the southern hemisphere (S).

HOTS

Critical Thinking

C-6.1

Why do you think the circles of latitudes get smaller as they approach the poles?

Latitudes numbered 1, 2 or 3 are close to the equator. Latitudes numbered 60, 70 or 80 are farther from the equator and closer to the poles. The latitude at each pole is numbered 90° N or 90° S. You can see the circles getting smaller in the diagram showing latitudes.

Which hemisphere are these two latitudes in?

20°N

40°S

Think and Answer

Critical Thinking

C-6.1

Why do the circles of latitude become smaller as they go from the equator to the poles?

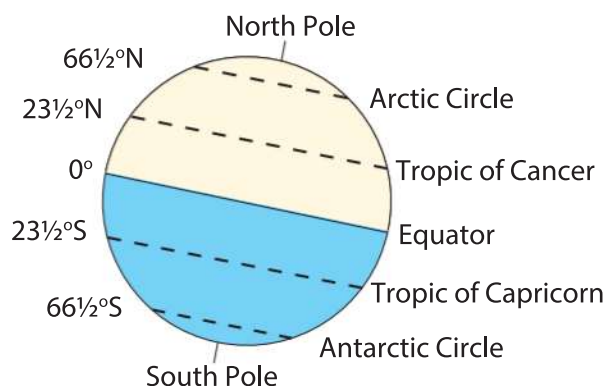
Special Latitudes

There are five special latitudes with the Equator at the centre. Two of them are in the northern hemisphere and the other two are in the southern hemisphere.

They are:

- The **Tropic** of Cancer at $23\frac{1}{2}^\circ\text{N}$
- The Arctic Circle at $66\frac{1}{2}^\circ\text{N}$
- The Tropic of Capricorn at $23\frac{1}{2}^\circ\text{S}$
- The Antarctic Circle at $66\frac{1}{2}^\circ\text{S}$

Locate them on the diagram given here.



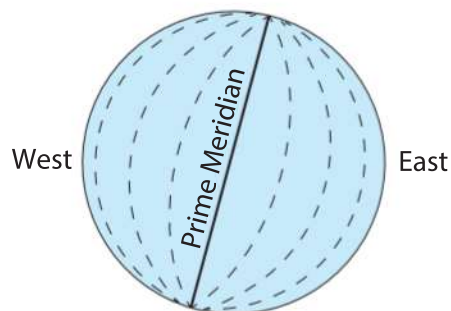
The special latitudes

LONGITUDES

Longitudes are lines that are drawn from one pole to the other. Each longitude starts at the North Pole and ends at the South Pole. Hence, each longitude forms a semicircle. You can see this in the diagram here.

The longitudes are also numbered. The special longitude in the centre is the **Prime Meridian** at 0° . Meridian means longitude. The other longitudes run East or West of the Prime Meridian, beginning and ending at the poles. Starting with Prime Meridian at 0° , the number of the longitude is followed by E (for East) or W (for West). A longitude that is 10°E is east of the prime meridian and 10°W is west of the prime meridian.

The longitudes continue east and west, round the globe, up to 180° . This is exactly opposite to the prime meridian on the other side of the globe. The eastern and western longitudes meet here at this longitude which is called the 180^th meridian. It does not have either E or W written after it. Can you say why this is so?



The lines of longitude



Knowledge of India

Earlier, many countries including India, had their own prime meridians. India's prime meridian passed through Avanti (the region near present-day Ujjain in Madhya Pradesh), which was known for astronomy. Now, the present-day standard meridian for India passes through Mirzapur, near Prayagraj in Uttar Pradesh, at $82\frac{1}{2}^\circ\text{E}$ which is known as the Indian Standard Time.

Eastern and Western Hemispheres

The two semicircles, the prime meridian and the 180^th meridian, form a circle that divides the earth into two halves—like an apple cut down the centre lengthwise. The eastern hemisphere lies to the east of the prime meridian and the western hemisphere lies to its west.

DIFFERENCES BETWEEN LATITUDES AND LONGITUDES

This table will help you understand the differences between these two important sets of imaginary lines.

Latitudes	Longitudes
1. Latitudes are circles of different sizes. The equator is the biggest circle and latitudes near the poles are the smallest circles. The latitude at each pole is a dot.	1. Longitudes are semicircles of the same size. They go from one pole to the other.
2. Latitudes are parallel to each other, so they never meet.	2. Longitudes meet at the poles. They are widest apart at the equator.
3. Latitudes run in an east-west direction.	3. Longitudes run in a north-south direction.

4. The equator is numbered as 0° . The other latitudes are numbered from 1° to 90° N or S, depending on which hemisphere they are in.

4. The prime meridian is numbered as 0° . The other longitudes are numbered from 1° onwards with E or W added to the number based on location. The 180° meridian where longitudes from the west and east meet, is on the opposite side of the 0° prime meridian.



A Better World

Sustainable Development Goals (SDGs)

Goal 16: Peace, Justice and Strong Institutions

Countries situated along the same longitudes and latitudes close to each other, are neighbours. It is important that neighbours cooperate and maintain peace and harmony amongst themselves.

What can we do to build harmony between neighbouring countries?

C-6.1

FINDING PLACES WITH THE HELP OF LATITUDES AND LONGITUDES

You have learnt earlier how to find places in a map. For this you need:

- a grid of numbered lines in each map
- an index where the names of places are given in alphabetical order.

In most maps, the lines of latitude and longitude form the grid. In atlases for young children the grids are numbered with an alphabet A, B, C and a number 1, 2, 3. In regular atlases, the grid numbers for a place will be the numbers (in degrees) of the latitude and the longitude with N, S, E or W after it. These are called the **coordinates** for the place.



Did You Know?

The 0° prime meridian at Greenwich, England, was adopted as the internationally recognized prime meridian at a meeting in 1884.



Discuss

Communication



Why are names given in an alphabetical order in an index?

C-6.1

What difference do you notice between the two indexes?

Latitudes and longitudes are useful not just in locating places. You will learn more about their uses in later grades.

NAMES	PAGE NO.	COUNTRY/REGION	GRID
A			
Abidjan	55	Cote d'Ivoire	B2
Abu Dhabi	51	UAE	D3
Abu Hills	8	Rajasthan	C4
Abuja	55	Nigeria	C2
Accra	55	Ghana	B2
Aconcagua Peak	58	Argentina	B3
Addis Ababa	55	Ethiopia	D2
Adelaide	63	Australia	C3
Adilabad	15	Telangana	B2
Adriatic Sea	60	Italy, etc.	C2
Afghanistan	51	Asia	E2
Agartala	13	Tripura	B3
Agra	12	Uttar Pradesh	A2
Ahmadabad	11	Gujarat	C3

Index of a junior school atlas

Names	Country/Region	P. No.	Lat.	Long.
A				
Aachen	Germany	81	50.47N	6.05E
Abadan	Iran	78	30.27N	48.25E
Abaetetuba	Brazil	97	1.45S	48.54W
Abapo	Bolivia	95	18.50S	63.27W
Abashiri	Japan	75	44.00N	144.15E
Abbottabad	Pakistan	14	34.09N	73.15E
Aberdeen	Scotland	83	57.09N	2.05W
Aberystwyth	Wales	83	52.24N	4.05W
Abha	Saudi Arabia	78	18.00N	42.34E
Abidjan	Cote d'Ivoire	87	5.26N	3.58W

Index of a middle school atlas





Values and Life Skills

- Learning how to locate places and following directions on a map are important skills to have. Which of these do you still have to learn?

1. Reading a map 2. Following directions on a digital or physical map 3. Locating places

REMEMBER

Locating Places on Earth

The lines of latitude and longitude are drawn on maps and globes but are imaginary lines on the surface of the earth.

Latitudes are drawn parallel to the equator.

The equator, the two tropics and the circles round the poles are all special latitudes.

Longitudes are semi-circles drawn from pole to pole.

The 0° Meridian and the 180° Meridian are special longitudes.

All the latitudes and longitudes are numbered.



NEW WORDS

Tropics	special latitudes near the equator
Prime	main
Meridian	a line that is drawn on a map from the North Pole to the South Pole
Coordinate	numbers or letters used to fix the position of a point on a map



EXERCISES

Remembering and Understanding

A Choose the correct answer.

1. Tropic of Cancer: $23\frac{1}{2}^{\circ}\text{S}/23\frac{1}{2}^{\circ}\text{W}/23\frac{1}{2}^{\circ}\text{E}/23\frac{1}{2}^{\circ}\text{N}$
2. Tropic of Capricorn: $23\frac{1}{2}^{\circ}\text{N}/23\frac{1}{2}^{\circ}\text{S}/23\frac{1}{2}^{\circ}\text{W}/23\frac{1}{2}^{\circ}\text{W}$
3. Antarctic Circle: $66\frac{1}{2}^{\circ}\text{N}/66\frac{1}{2}^{\circ}\text{S}/66\frac{1}{2}^{\circ}\text{W}/66\frac{1}{2}^{\circ}\text{E}$
4. Read the statements below and choose the correct option.
Statement (A): There are five special latitudes.
Statement (B): These are all in the Northern Hemisphere.
 (a) Both A and B are true, and B is the correct explanation of A.
 (b) Both A and B are true, but B is not the correct explanation of A.
 (c) A is true but B is false.
 (d) A is false but B is true.

B. Fill in the blanks.

1. The equator is numbered as
2. The North Pole is numbered as and the South Pole as
3. The prime meridian is numbered as
4. The meridian opposite the prime meridian is numbered as
5. The that help us to find a place in a map are called coordinates.

C. Put these in the right columns.

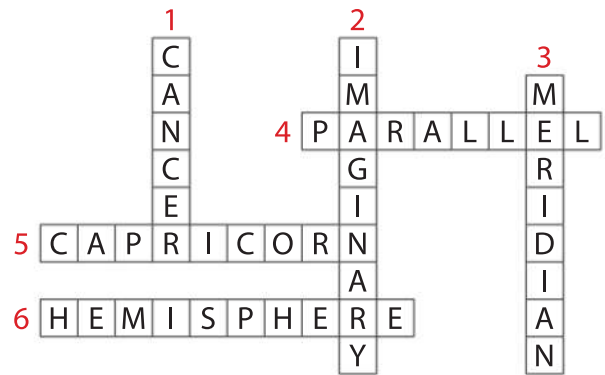
Make two columns in your notebook—latitudes and longitudes. Put the following words and numbers in the right columns:

circles semi-circles parallel meet at the poles 90°N
90°S 180th meridian equator prime meridian

D. Write the clues to this crossword in your notebook.

E. Answer the following questions.

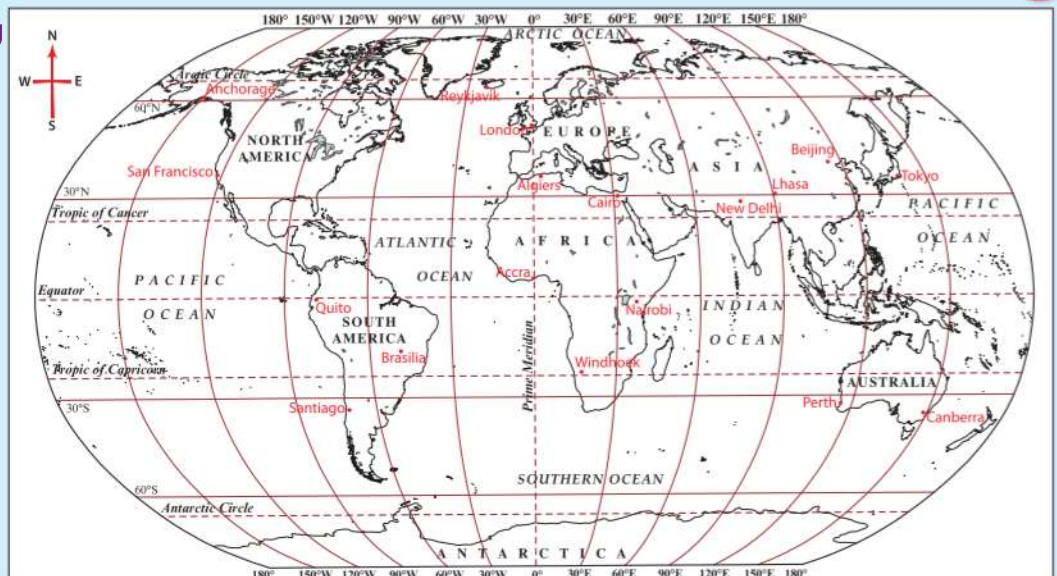
1. What are the two sets of lines used to locate places on maps and globes called?
2. Explain why these lines are called imaginary lines?
3. Which are the lines that run parallel to the equator?
4. Name the lines that meet at the poles.
5. What is the longitude that is exactly opposite the prime meridian called?



DO AND LEARN

A. Map Reading

TP



Look at the map on page 83 and name the following. Take 1 point for each correct answer.
Which are the following:

1. The latitude just north of New Delhi _____
2. The continents that the Tropic of Cancer passes through _____
3. The continents that the Tropic of Capricorn passes through _____
4. Two places that are on or almost on the equator _____
5. Three places that are just south of 30°S _____
6. Two places that are north of the Tropic of Capricorn _____
7. Two places located between the Arctic Circle and 60°N _____
8. Two places that are almost on the prime meridian _____
9. Two places that are just west of 150°E _____
10. Two places that are west of 120°E _____

C-5.3

Map Reading

B. On Your Own

Here is a sentence to help you remember the sequence of the tropics –
North — Tropic of Cancer, South — Tropic of Capricorn: NTCaSTCap

The sentence: Naughty Tiger Can't Swallow Tiny Capsules.

Make another sentence of your own that will help you remember.

C-3.3

Creativity

SI

English

C. In a Group

Play an atlas game with a partner. Both of you should have the same atlas. Take turns to do the following:

1. The first person names a place from the index. The second person has to find it in the index, **calls out the coordinates** and then locate it in the map.
2. The first person counts to 20, slowly. Give two points if the place is found before the count is finished. **The first person with the highest points is the winner.**

C-5.2

Collaboration

D. Pair and Share

How would you find your way back to your hotel if you were lost during a holiday?

Share your ideas with your partner.



Weblink For more information, go to:

- **Longitude and Latitude:** <https://www.youtube.com/watch?v=cTrsvGytGG0> (Accessed on 31 July 2025)



Self-review

What I Learnt

Colour the smileys to show how well you have understood the concepts.	Completely	Mostly	Need help
• I can explain what latitudes and longitudes are.			
• I can distinguish between latitudes and longitudes and elaborate on their features.			
• I can use latitudes and longitudes to locate places on a map.			